SOURCE

WHERE SOLUTION BEGINS.

ABOUT THE SYSTEM

AI-Drophonics is an IoT-based hydroponics monitoring and automation system with integrated AI crop disease detection, purpose-built for urban farms in the Philippines.

Built by SOURCE for Gulayan at Bulaklakan Integrated National Urban Farm in Quezon City, the system helps farmers catch crop disease before it spreads without watching the farm all day.

HOW IT WORKS

A Raspberry Pi 5 edge device runs real-time YOLOv8n object detection and receives readings from environmental sensors. Sensor-driven responses automate farm actions, while the companion Android app brings farm status and alerts to the farmer.

MONITOR, DETECT, ACT

The app provides real-time pH, temperature, humidity, and water-level visibility; push notifications for disease detection; AI-generated treatment recommendations; and a centralized farm dashboard.

PROJECT TEAM

Concepcion, Justine Altaire B. | De Juan, Axel P. | Lingatong, Nazh Manuel V. | Odon, Marc Joseph V. | Salvador, Gerard Jose N.

Adviser: Maria Carolina Bautista | Subject Teacher: Arnel Balasta

ACADEMIC YEAR 2026-2027 / QUEZON CITY, PHILIPPINES